

Homework



Homework

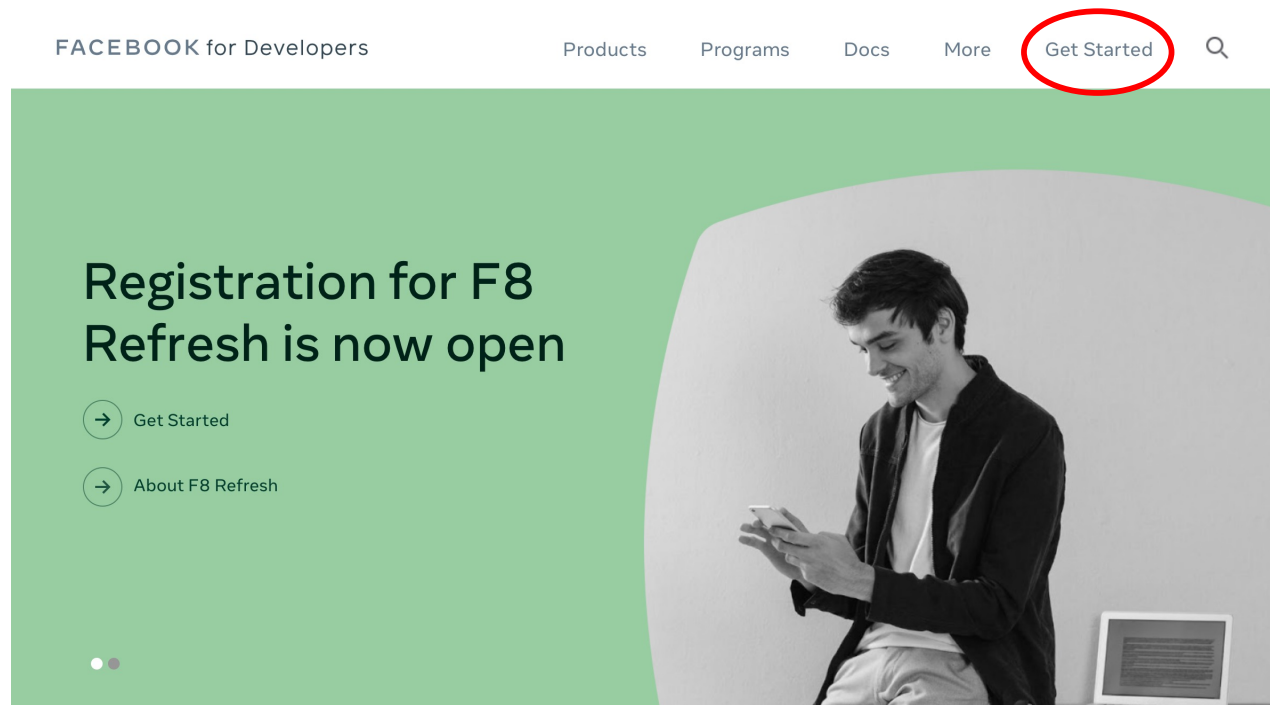
- In today's homework, you will implement OAuth Facebook Login API onto a React app.
- These are the steps:
 1. Set up a Facebook App
 2. Create React.js App
 3. Install Package "react-facebook-login"
 4. Adding Facebook login to our React app
 5. Run React App that shows option for Facebook Login
 6. Redirect to App after login with Facebook



Homework

Set up a Facebook App

- You will first need to register as a Facebook App Developer. Head to <https://developers.facebook.com/apps/> to register. You will be asked to login to your Facebook if you are not already logged in.
- Once you are redirected to the Facebook for Developers home page, click "Get Started":





Homework

Set up a Facebook App

- You will be taken to the register information screen. Click “Continue”.
- Next, on the Verify account screen, enter your phone number and click Send to get a verification code.

Verify Your Account

Verify your developer account by adding a mobile number.

Country
United States (+1) ▼

Mobile number
Enter your phone number

This number will be saved to your Facebook profile. We use mobile numbers to send SMS notifications, help you log in, and personalize experiences, like connecting people and improving ads for everyone on our products. Only you will see your mobile number on your profile. [Learn More](#)

[Send Verification SMS](#)

You can also verify your account by [adding a credit card](#)



Homework

Set up a Facebook App

- On the contact info screen, enter your email and click on the agree checkbox. Click Send or Continue (whichever is the case).

The screenshot shows the Facebook app setup process. On the left, a vertical progress bar has four steps: 'Register' (checked), 'Verify accou...' (checked), 'Contact info' (selected with a blue circle), and 'About you' (not selected). The main area is titled 'Enter Your Preferred Contact Email'. It contains a paragraph explaining that the email will be added to the Facebook account for notifications and personalization. Below this is a 'Learn More' link. The 'Primary email' section has a text input field with the placeholder 'Enter your email'. Below the input field is an unchecked checkbox with the text: 'I agree to receive marketing-related electronic communications from Facebook, including developer news, updates and promotional emails. (You may unsubscribe from these emails at any time by clicking unsubscribe at the bottom of the email. You can also update your email preferences in Developer Settings.)'. At the bottom right is a blue button labeled 'Send Verification Email'.



Homework

Set up a Facebook App

- On the About you screen, choose “Developer” and the click “Complete Registration”

Which of the following best describes you?

Help us improve your experience by telling us which of the following roles best describe you.

	Developer ☒		Marketer ☐
	Analyst ☐		Product Manager ☐
	Student ☐		Owner/Founder ☐
		Other ☐	

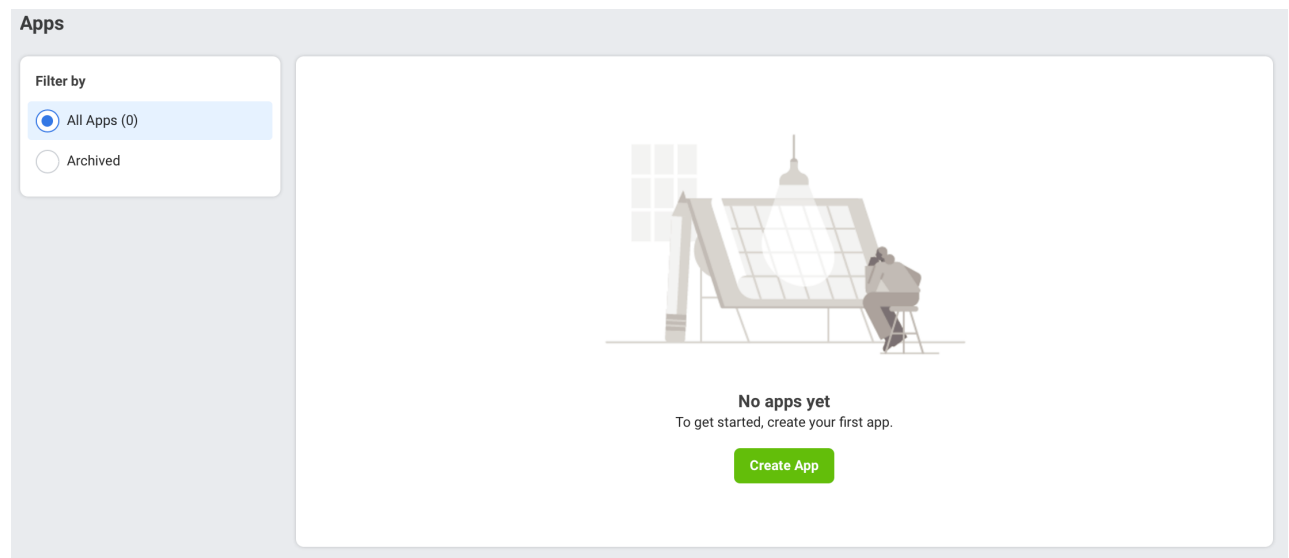
Complete Registration



Homework

Set up a Facebook App

- You will be redirected to the dashboard. Then click “Create App”.

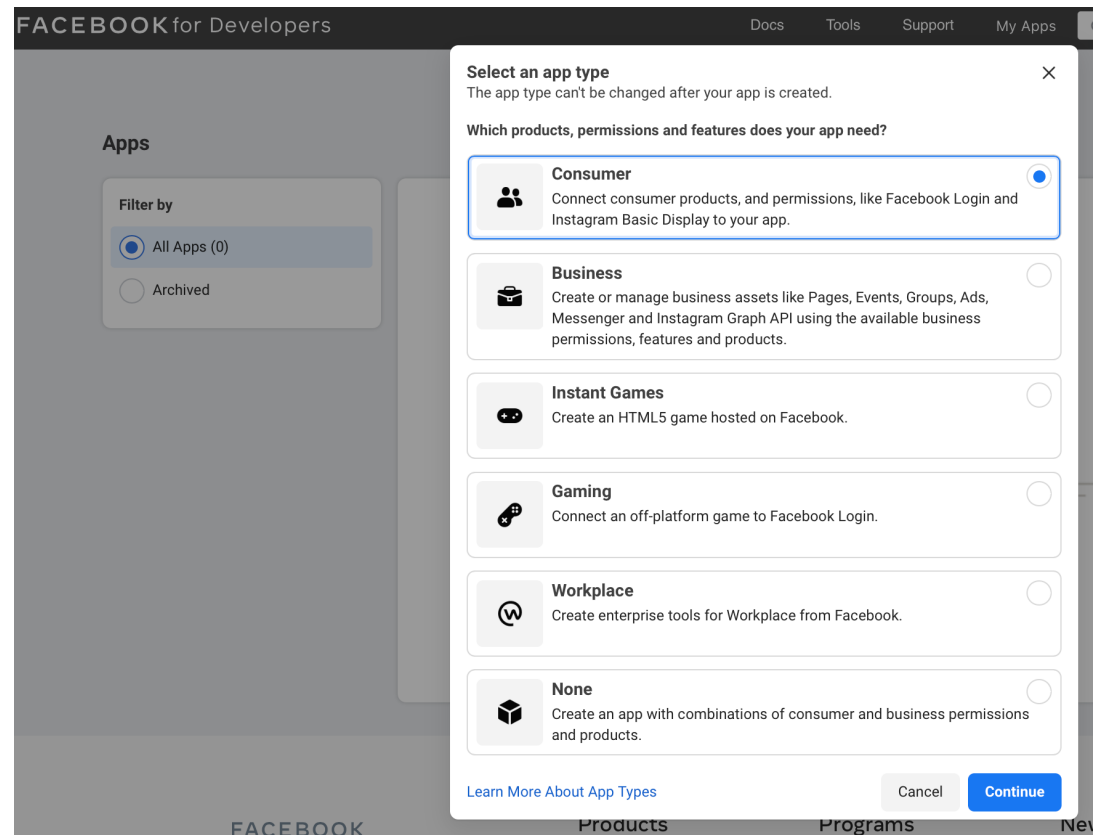




Homework

Set up a Facebook App

- A pop up screen appears to select the type of app. Select the first option – consumer. Then click continue.





Homework

Set up a Facebook App

- In the following screen, fill the App Display Name with the name of your app and your email that was previously registered. For our app, use the name “React_Login”. After that, click “Create App”.

The screenshot shows the Facebook App creation interface. On the left, under the heading "Apps", there is a "Filter by" section with two options: "All Apps (0)" (selected with a blue radio button) and "Archived" (unselected with a grey radio button). On the right, a modal window titled "Create an App" is open. It contains the following fields and text:

- App Display Name**: "This is the app name associated with your app ID." The input field contains "React_Login".
- App Contact Email**: "This email address is used to contact you about potential policy violations, app restrictions or steps to recover the app if it's been deleted or compromised." The input field is empty.
- Do you have a Business Manager account? · Optional**: "In order to access certain aspects of the Facebook platform, apps may need to be connected to a verified Business Manager account. If you haven't yet set up an account, you can create one now or later in the process." A dropdown menu shows "No Business Manager Account selected".
- At the bottom, it says "By proceeding, you agree to the [Facebook Platform Terms](#) and [Developer Policies](#)".
- There are two buttons at the bottom: a grey "< Back" button and a blue "Create App" button.



Homework

Set up a Facebook App

- After checking the captcha dialog and click submit, you will be redirected to the Facebook Application Dashboard.
- Your Facebook app is now set up.
- Next – create the React app.

The screenshot shows the Facebook for Developers dashboard for an app named 'React_Login' with App ID 149676430506881. The app is in development and is a consumer app. The dashboard features a sidebar with navigation links: Dashboard, Settings, Roles, Alerts, and App Review. The main content area is titled 'Add Products to Your App' and lists six products that can be added to the app: Analytics, Audience Network, Facebook Login, Instagram Basic Display, oEmbed, and Webhooks. Each product card includes a brief description and a 'Set Up' button.

Product	Description	Action
Analytics	Understand how people engage with your business across apps, devices, platforms and websites.	Read Docs / Set Up
Audience Network	Monetize your app and grow revenue with ads from Facebook advertisers.	Read Docs / Set Up
Facebook Login	The world's number one social login product.	Read Docs / Set Up
Instagram Basic Display	The Instagram Basic Display API allows users of your app to get basic profile information, photos, and videos in their Instagram accounts.	Read Docs / Set Up
oEmbed	Embed Facebook and Instagram posts, Pages and videos in websites and apps.	Read Docs / Set Up
Webhooks	Subscribe to changes and receive updates in real time without calling the API.	Read Docs / Set Up



Homework

Create React.js App

- First, create a new repository **m5-w3-d1-homework** in your remote Github and then clone this in your WESTCLIFF folder.
- Launch VS Code and navigate to this clone folder.
- On a bash terminal, create a React app:
npx create-react-app react-fb_login
- Next, CD to **react-fb_login**.
- Then **npm start** to test run the default app. If things work as expected, proceed to the next step.



Homework

Install the “ react-facebook-login ” library

- You will install this library next so you can use the module.
- `npm install react-facebook-login`
- After install is complete, check the `package.json` file to ensure `react-facebook-login` has been added as dependencies.

```
{
  "name": "react-fb_login",
  "version": "0.1.0",
  "private": true,
  "dependencies": {
    "@testing-library/jest-dom": "^5.12.0",
    "@testing-library/react": "^11.2.7",
    "@testing-library/user-event": "^12.8.3",
    "react": "^17.0.2",
    "react-dom": "^17.0.2",
    "react-facebook-login": "^4.1.1",
    "react-scripts": "4.0.3",
    "web-vitals": "^1.1.2"
  },
}
```



Homework

Set React App to run on HTTPS

- For secure login, it is recommended that you set up your React app project to run with HTTPS. Facebook technically requires all their logins to use HTTPS. To set this up, replace the script of "start" in the package.json file with:

"start": "**HTTPS=true** react-scripts start",

```
"scripts": {  
  "start": "HTTPS=true react-scripts start",  
  "build": "react-scripts build",  
  "test": "react-scripts test",  
  "eject": "react-scripts eject"  
},
```



Homework

Integrate Facebook Login Button

- First, install react-bootstrap and bootstrap 4:

npm install react-bootstrap bootstrap

- Open **App.js** and remove all the default codes in there.
- Now add the following imports:

```
1  import React, { useState } from 'react';
2  import FacebookLogin from 'react-facebook-login';
3  import { Card } from 'react-bootstrap';
4  import "bootstrap/dist/css/bootstrap.min.css";
5  import './App.css';
```



Homework

Integrate Facebook Login Button

- Next, create the App component with a few useState setups and in the return, a container:

```
function App() {  
  const [login, setLogin] = useState(false); //set up login  
  const [data, setData] = useState({}); //set up fb data  
  const [picture, setPicture] = useState(''); //set up fb profile image  
  
  return (  
    <div class="container">  
  
    </div>  
  )  
}
```



Homework

Integrate Facebook Login Button

- Add a React-Bootstrap card –

```
<div class="container">
  <Card>
    <Card.Header></Card.Header>
    <Card.Body>
      <Card.Text></Card.Text>
    </Card.Body>
  </Card>
</div>
```

- Before proceeding, let's breakdown the logistic of this app:
 - Upon app loads:
 - user has 2 login options: standard site login and FB login (you will only work on the FB login in this homework).
 - After FB login, user will be redirected to the home component.



Homework

Integrate Facebook Login Button

- Within the card, add the following:

```
<Card.Header>
|   <h1>My React App</h1>
</Card.Header>
<Card.Body>
|   <Card.Text>
|       <h3>Please login using one of the following:</h3>
|       { /* Login Form */ }
|       { /* FB Login Button */ }
|   </Card.Text>
</Card.Body>
```



Homework

Integrate Facebook Login Button

- Let's create the Login Form component (add this below the App component):

```
function LoginForm() {  
  return(  
    <form className="border mt-3 mb-5 p-3 bg-white">  
      <label className="m-2">Name:</label>  
      <input type="text" name="name" placeholder="Your name" />  
      <label className="m-2">Email:</label>  
      <input type="email" name="email" placeholder="Your Email" />  
      <input type="submit" value="Login" className="btn bg-success text-white my-3"/>  
    </form>  
  )  
}
```



Homework

Integrate Facebook Login Button

- Now you can call this Login Form component in the App return:

```
<Card.Text>
  <h3>Please login using one of the following:</h3>
  { /* Login Form */ }
  <LoginForm />
  { /* FB Login Button */ }
</Card.Text>
```

- Next – the FB login button:

For more React Facebook button:

<https://www.npmjs.com/package/react-facebook-login>

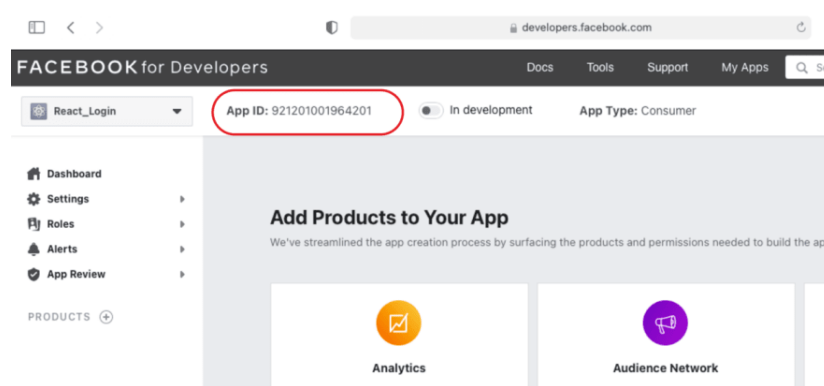
```
{ /* FB Login Button */ }
<FacebookLogin
  appId="1088597931155576"
  autoLoad={false}
  fields="name,email,picture"
  scope="public_profile,user_friends"
  callback={responseFacebook}
  icon="fa-facebook"
/>
</Card.Text>
```



Homework

Integrate Facebook Login Button

- Return to your Facebook for Developers account to get your App ID:



- Copy this ID and paste to replace it in App.js:

```
<FacebookLogin
  appId="1088597931155576"
  autoLoad={false}
  fields="name,email,picture"
  scope="public_profile,user_friends"
  callback={responseFacebook}
  icon="fa-facebook"
/>
```



Homework

Integrate Facebook Login Button – Some Information

- The FacebookLogin component is a button-like component. The FacebookLogin button will grant access to your FB name, email, profile pic and so on when you successfully logged in.
- In the button, there's a callback to a handler to capture the response from the FacebookLogin component. In this response you will get an access token and also some profile data from your Facebook.



Homework

How the FacebookLogin library works

- To understand how the FacebookLogin library works, we need to take a look at the code we have installed in our project. Open the react-facebook-login folder in the directory `"/ node_modules / react-facebook-login /"`. Then open the facebook.js file in the directory `"/node_modules/react-facebook-login/dist/facebook.js"`. Before the FacebookLogin component is rendered, the attributes given to this component will be fetched and processed by FB SDK, especially in the appId we prepared earlier.

```
172 window.fbAsyncInit = function () {  
173   window.FB.init({  
174     version: 'v' + version,  
175     appId: appId,  
176     xfbml: xfbml,  
177     cookie: cookie  
178   });
```



Homework

How the FacebookLogin library works

- The login function is executed when we click the login button. Then the response will be caught by the checkLoginState function.

```
123     if (_this.props.isMobile && !disableMobileRedirect) {  
124         window.location.href = '//www.facebook.com/dialog/oauth?' + (0, _  
125     } else {  
126         window.FB.login(_this.checkLoginState, { scope: scope, auth_type:  
127     }
```



Homework

How the FacebookLogin library works

- The authResponse we get will then be processed by the responseApi function.
- The responseApi data will be returned to the callback function which in our code was previously used to get the profile data response from our Facebook account.

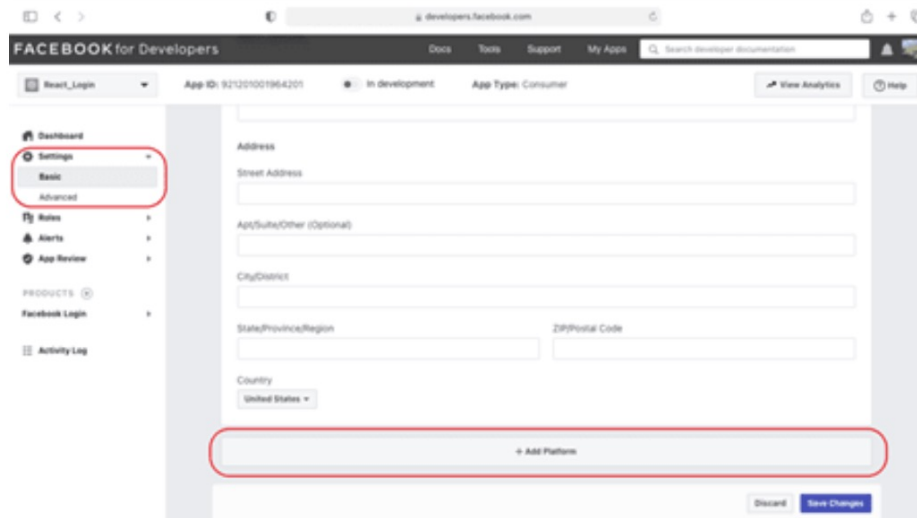
```
69     }, _this.responseApi = function (authResponse) {
70         window.FB.api('/me', { locale: _this.props.language, fields:
71             _extends(me, authResponse);
72             _this.props.callback(me);
73         });
74     }, _this.checkLoginState = function (response) {
75         _this.setStateIfMounted({ isProcessing: false });
76         if (response.authResponse) {
77             _this.responseApi(response.authResponse);
78         } else {
79             if (_this.props.callback) {
80                 _this.props.callback({ status: response.status });
81             }
82         }
83     }
```




Homework

Integrate Facebook Login Button

- Now that you have a little better understanding how the FacebookLogin library works, you can continue to the next step.
- Open your Facebook Developer dashboard, click on **Settings** on the left navigation, and then select **Basic**. Continue scrolling down until you find the **+ Add Platform** button.

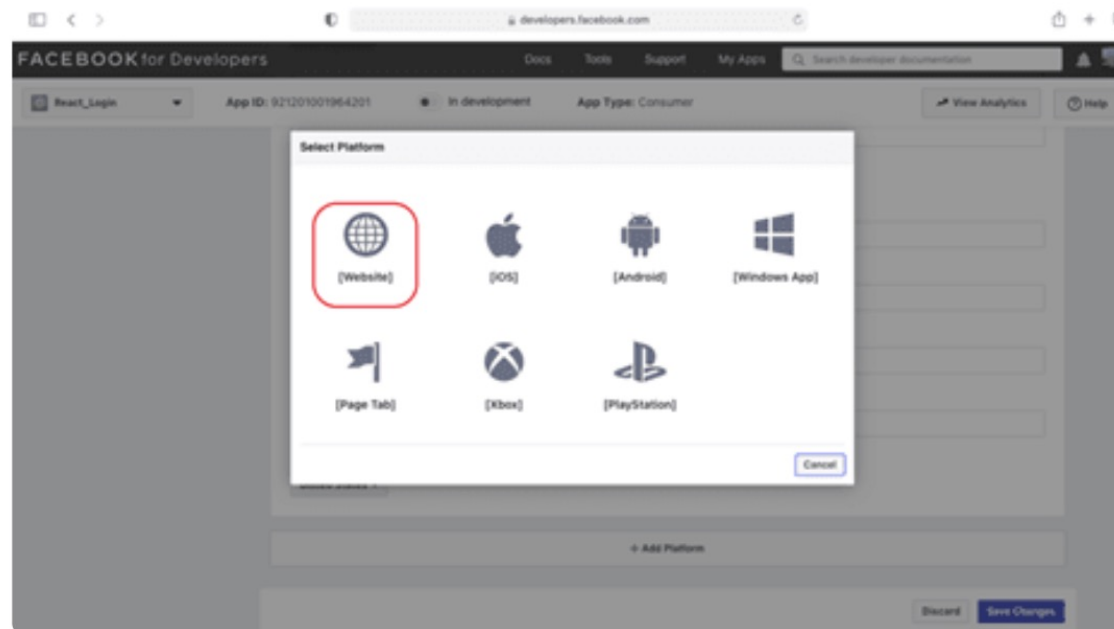




Homework

Integrate Facebook Login Button

- A pop up window will appear next. Select **Website** as the platform:





Homework

Integrate Facebook Login Button

- After that you will be taken to the following screen. Fill in the **Site URL** container with the network URL. You can obtain this in the bash terminal after you'd entered `npm start`. An example URL: <http://127.0.0.1:3000/auth/facebook/callback> and then click **Save Changes**.

The screenshot shows the Facebook for Developers dashboard for an app named 'React_Login'. The app ID is 921201001964201, and it is in development mode. The 'Basic' settings tab is selected. The 'Website' section is highlighted with a red box, showing the 'Site URL' field with the example URL 'http://127.0.0.1:3000/auth/facebook/callback'. Other fields include 'City/District', 'State/Province/Region', 'ZIP/Postal Code', and 'Country' (set to United States). The 'Save Changes' button is visible at the bottom right.



Homework

Integrate Facebook Login Button

- Next, let's return to App.js. You will need to wrap all these JSX elements with a wrapper. So far, you have been using a div element as the wrapper. There are others besides the standard HTML fare. These ones will not add an unnecessary node on the real/browser DOM:
 - `<React.Fragment></React.Fragment>`
 - `<></>` (shorthand version)

```
<React.Fragment>
  <h3>Please login using one of the following:</h3>
  { /* Login Form */ }
  <LoginForm />
  { /* FB Login Button */ }
  <FacebookLogin
    appId="1088597931155576"
    autoLoad={false}
    fields="name,email,picture"
    scope="public_profile,user_friends"
    callback={responseFacebook}
    icon="fa-facebook"
  />
</React.Fragment>
```



Homework

Integrate Facebook Login Button

- The callback in the FB button calls a default handler **responseFacebook**. You will create this handler between the `useState()` and `return()`. The basic handler is mentioned in the `react-fb-button` documentation.
- On this handler, the `useState` methods are added to handle the response from FB. In addition to that, if an access token is sent with the response, the login method is set to `true`, otherwise `false`. These conditions will be used to determine what content/components get to display on the app screen.

```
const responseFacebook = (response) => {  
  console.log(response);  
  setData(response);  
  setPicture(response.picture.data.url);  
  if (response.accessToken) {  
    setLogin(true);  
  } else {  
    setLogin(false);  
  }  
}
```



Homework

Integrate Facebook Login Button

- If you recall earlier, our app logic is when app first loads (not login), the screen will display the login form and FB login button. Once logged in using FB, the home component will be displayed. Let's start with not login:

```
<Card.Text>
  {!login &&
    <React.Fragment>
      <h3>Please login using one of the following:</h3>
      {/* Login Form */}
      <LoginForm />
      {/* FB Login Button */}
      <FacebookLogin
        appId="1088597931155576"
        autoLoad={false}
        fields="name,email,picture"
        scope="public_profile,user_friends"
        callback={responseFacebook}
        icon="fa-facebook"
      />
    </React.Fragment>
  }
</Card.Text>
```



Homework

Integrate Facebook Login Button

- And now the successful login.
When that happens, the home component will be called. Along with it, FB data will be passed to it via attributes.

```
        icon="fa-facebook"
      />
    </React.Fragment>
  }

  {login &&
    <Home fbpic={picture} fbdata={data} />
  }
</Card.Text>
```



Homework

Home Component

- Next, you will create the home component. Create this below the Login Form component:

Note: notice the two parameters instead of the standard props. You can de-structure the props by putting in the attributes within a set of curly braces. In that case, you do not need to use the property as `props.fbpic` but just `fbpic`.

- And finally, export the App component.

```
function Home({fbpic,fbdata}) {  
  return(  
    <React.Fragment>  
      <img src={fbpic} alt={fbdata.name} />  
      <h3 className="d-inline text-success mx-2">  
        Welcome back {fbdata.name}!  
      </h3>  
      <p className="my-5">This is the home page of the app.</p>  
    </React.Fragment>  
  )  
}  
  
export default App;
```




Homework

A little styling

- Add the following styling:

```
<Card style={{ width: '800px' }} className="mx-auto mt-5">
  <Card.Header className="pb-4">
    <h1>My React App</h1>
  </Card.Header>
```

- And on **App.css**:

```
# App.css > ...
.card-text form {
  width: 250px;
}
```

Run your React App to login with Facebook

- If your app is not already running on the browser, **npm start** on your bash terminal.



Homework

- What the app looks like. Click the **FB Login button**.

My React App

Please login using one of the following:

Name:

Email:

 LOGIN WITH FACEBOOK



Homework

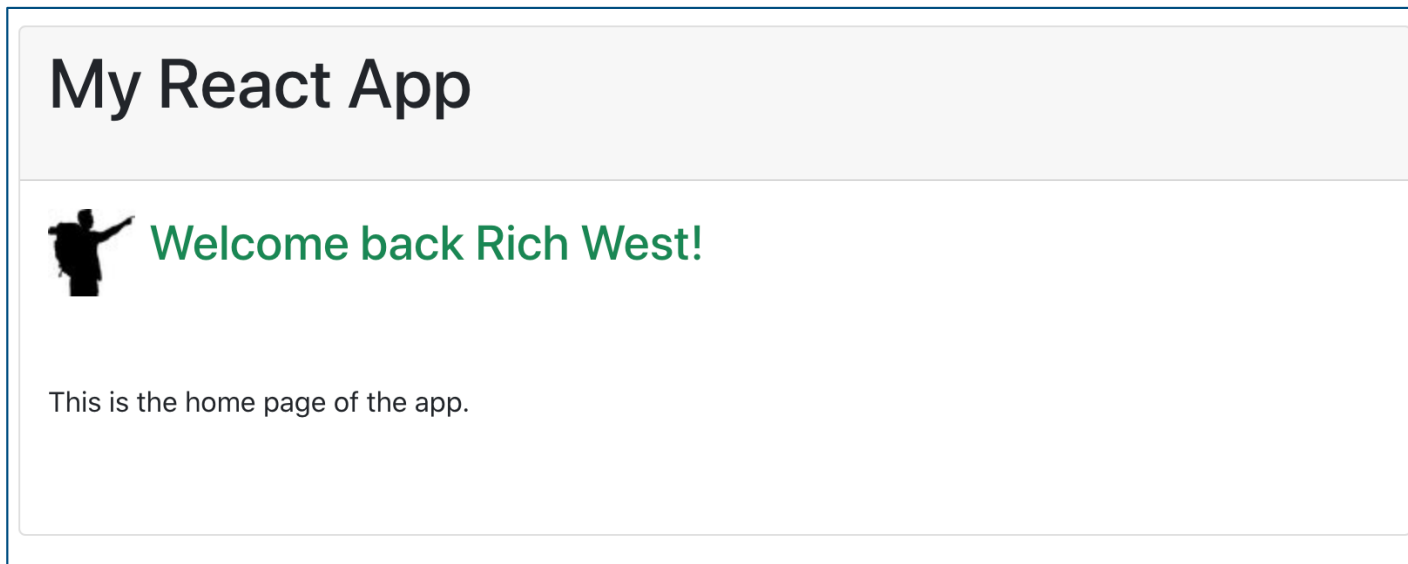
- The FB login will pop up. Enter your FB password and click Log In.

The image shows a screenshot of a web application with a Facebook login pop-up. In the background, a form titled "My React" asks for a name and email, with a green "Login" button. A blue button at the bottom says "f LOGIN WITH FACEBOOK". The foreground pop-up is a Facebook login window titled "Facebook" with the URL "facebook.com/login.php?skip_api_login=1&api_key=149676430506881&kid_d...". It prompts the user to "Log in to use your Facebook account with React_Login." and contains fields for "Email or Phone:" (with "richloke@westcliff.edu" entered) and "Password:". Below these fields are buttons for "Log In" (blue) and "Create New Account" (green), along with a link for "Forgot account?".



Homework

- After authorization is successful, you will be taken to the app's home screen:





Homework

DUE:

On or before this Sunday, 10.30PM PT

Submission:

Post GitHub URL on GAP Week 3 Day 1 Homework

The background is a solid light blue color. In the bottom-left corner, there is a decorative geometric pattern consisting of several overlapping triangles and squares in different shades of blue, ranging from a very dark navy blue to a medium blue.

Questions?



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